Manuscript Terminology Consistency Plan

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A plan to align the terminology in all `report.Rmd` and `report-slim.Rmd` manuscripts with the project lexicon (`docs/29-nof1-precision-medicine-lexicon.md`) and with each other.

Execution status (2026-07-30)

Executed with D1 locked to US English. Tier 1 was applied to all 21 `report*.Rmd` files (the plan was written against 13; six variants have been added since, and paper 11 did not exist). Tier 2, Tier 3, and the Section 5 full-versus-slim pass were applied where the targets still existed; most had already converged through intervening revisions. Section 6 items remain open and are for the author.

What changed, and what was already clean, is recorded at the end of this file. All 21 manuscripts were re-rendered and build.

Scope and method

Thirteen manuscript files across ten papers were reviewed against the lexicon (prose, headings, captions, and abstracts only; R code chunks and in-model variable names such as `c.bm`, `Dbc`, `carryover_t1half` were excluded). Papers 01, 02, and 06 have both a full and a slim variant; the variants were reviewed together for cross-file consistency. The `rgt` paragraph blocks are still Lorem ipsum placeholders and were not reviewable; a second pass will be required once the author writes that prose.

Finding in one line. No statistical term is *wrong*; the issues are (i) inconsistent surface forms for the same concept across and within papers, (ii) acronyms used before first-use expansion, and (iii) a few colloquial coinages and one genuinely narrow lexicon term ("washout") used loosely. Three *substantive* (non-terminological) issues surfaced incidentally and are listed separately in Section 6; they require author judgement and must not be folded into a mechanical edit.

Important execution constraint. These files live under `~/Library/CloudStorage/` (Dropbox). Per the project guardrails, in-place stream editors (`sed -i`, `perl -i`, etc.) must NOT be used on cloud-mounted paths. All edits must go through the Edit tool (which is path-aware) or the copy-to-`/tmp`, edit, `mv`-back pattern. Global substitutions should be applied per file with `replace_all`, never with a recursive `sed -i`.

1. Decisions to lock before editing

These choices propagate everywhere, so confirm them first. Recommended defaults are evidence-based (current corpus counts in parentheses).

#	Decision	Recommended default	Evidence
D1	Spelling convention	**`US / -ize, -or, modeling`** (locked 2026-07-30; overrides the frequency-based recommendation below)	The corpus leans British (modelling 73 vs modeling 43; characterise 26; behaviour 18 vs 3; generalised 16 vs 8), but frequency records drift, not a decision. The project-wide standard is US English, so the direction of travel is British -> US.
D2	Casing of "Type I error"	`Type I error` (cap T, cap I, no hyphen)	Type I error 121 vs type I 61 vs type-I 45
D3	Biomarker-interaction surface form	`biomarker-treatment interaction`	91 vs biomarker-by-treatment 45; matches lexicon §10 headword
D4	Monte Carlo SE form	spell `Monte Carlo standard error (MCSE)` at first use, then `MCSE`	MCSE 281, MC SE 59, spelled 57, "Monte Carlo SE" 25
D5	"washout" usage	reserve `washout` for an inserted treatment-free clearance gap; use `off-drug period / occasions` for the general post-discontinuation span	off-drug 279 vs washout 92; the papers' own thesis is that carryover contaminates these spans, so "washout" is self-contradictory there (lexicon §2)

#	Decision	Recommended default	Evidence
D6	Precision vs personalised medicine	`precision medicine` (primary), "personalised medicine" only as a glossed synonym	precision 11 vs personalised 4; lexicon §10 headword
D7	Participant unit noun	`participant` in prose; use "subject"/"cluster" only where theory requires (e.g. sandwich estimator)	papers 05/08 use participant; 10 mixes cluster/subject/participant

2. Tier 1 - project-wide standardisations (safe, high value)

Apply across all 13 files. Each is a low-risk, presentational fix that does not alter meaning. Apply with the Edit tool (`replace_all` per file), checking each hit in context (especially D5, which is contextual, not a blind replace).

1. **Type I / Type II error casing** (D2). Normalise `type I error`, `type-I error`, `type-I`, `Type-I` -> `Type I error`; likewise `Type II`. Affects 02, 04, 05, 07, 08, 09, 10. (~106 non-canonical instances.)
2. **Biomarker-treatment interaction** (D3). `biomarker-by-treatment interaction` -> `biomarker-treatment interaction`; also normalise `treatment-by-biomarker`, `biomarker:treatment`, "predictive interaction" where they denote the same estimand. Affects 02, 03, 06, 08, 09, 10. (~45 instances.)
3. **MCSE** (D4). `MC SE` and `Monte Carlo SE` -> `MCSE`; ensure one spelled-out first use per file. Affects 01, 02, 07 (and any ADEMP paper). (~84 instances.)
4. **HTE first-use + label**. Spell `heterogeneity of treatment effects (HTE)` at first use in every paper that uses the concept, then `HTE`. Replace loose paraphrases ("treatment-response heterogeneity", "individual treatment heterogeneity", "between-person response variability") with HTE or with "patient-by-treatment interaction" where the variance-component meaning is intended. Critical in 06 (concept central but acronym never introduced) and 01 (used before expansion). Also 03, 04.
5. **"washout" -> "off-drug period/occasions"** (D5). Contextual: change only where the text means the post-discontinuation span in which carryover decays; keep "washout" where a clearance gap is genuinely inserted. Affects 01, 02, 04. (~92 "washout" hits to triage.)
6. **Acronym first-use expansion.** Ensure each is spelled out at

first reader-facing use, then abbreviated: `DGP` (data-generating process/mechanism), `MVN` (multivariate normal), `LME` (linear mixed-effects model), `MCAR/MAR/MNAR`, `OL/BDC/OL+BDC`, `CO` (crossover), `PK/PD`, `BLUP`, `SEM`, `IL-6`, `RM-ANOVA`, `OFAT`. The three-part `bullets/orig` structure aggravates this because a `bullets` block often leads with a bare acronym that the later `orig` block expands; fix by expanding in whichever block appears first.

7. **Spelling normalization** (D1, direction reversed on 2026-07-30). `generalised` -> `generalized`, `modelling` -> `modeling`, `behaviour` -> `behavior`, `parameterise` -> `parameterize`, `characterise` -> `characterize`, `normalise` -> `normalize`, `whilst` -> `while`, `artefact` -> `artifact`, `grey` -> `gray`, etc. Preserve British spelling inside verbatim quotations and in published citation or reference titles. Do not touch the US-English false friends (`exercise`, `revise`, `raise`, `advise`, `promise`, `compromise`, `analyses`, `characteristic`, `programmer`).
8. **Code tokens leaking into prose.** Replace reader-facing code identifiers with their prose terms: `OLBDC` -> `OL+BDC` (02), `modGompertz` -> "modified Gompertz function" in narrative (07; keep `modGompertz` only when referring to the R function). Keep all genuine model-symbol references (`$D_{bc}$`, `$c.bm$`, `bm:Dbc`) as code.

3. Tier 2 - per-paper consistency fixes

Paper	Fix	Notes
01	Standardise the drug-indicator naming ("treatment indicator" / "drug indicator" / "drug exposure variable" -> one prose term for the DGP indicator and one for the analysis variable).	Within-paper inconsistency for `$D_{it}$` / `Dbc`.

Paper	Fix	Notes
02	Settle on ONE label for analysis spec A2 (“matched-decay” / “exposure-weighted” / “continuous-D_{bc}”); reserve “matched-decay” for the matched DGP cell. Fix “censored” where it means dropped/missing (lexicon §8 vs §11). Standardise “Hybrid N-of-1 design” and “crossover (CO)” (drop “classical/traditional” alternation).	A2 label multiplicity (~25) is the biggest clarity win.
03	Disambiguate unobserved-group vocabulary: use `latent class` for unobserved groups; reserve `subgroup`/`stratum` for observed splits (lexicon §10). Gloss `subadditivity` and `frailty` at first use; change “diagnostic biomarker” -> `predictive biomarker` where differential benefit is meant.	“frailty” is an unglossed third synonym for random effect.

Paper	Fix	Notes
06	Replace colloquial `lumped` -> `one-component` / `single-indicator` (both already used). Define `response expectancy` once, then use "expectancy" rather than rotating belief/suggestion/suggestibility/optimism. Gloss `balanced-placebo (open-hidden) design` at first use. Standardise "between-participant".	See also the Study B/C cross-file bug in Section 6.
07	Capitalise `Gompertz` consistently in prose (lowercase only as a factor level/code). Remove `modGompertz` from narrative (Tier 1.8). Add a one-line gloss tying Hill / sigmoid-Emax / indirect-response to the lexicon "Emax / dose-response model" cluster.	Mostly cosmetic but pervasive.
08	Disambiguate `main effect`: use `average treatment effect (ATE)` for the population mean effect; keep "main effect" only for the ANOVA-factor sense. Introduce `anti-conservative (inflated Type I error)` once, then use one term.	"main effect" overload is a genuine ambiguity.

Paper	Fix	Notes
09	Consolidate `informative dropout` (canonical) over "biased dropout"/"attrition" (keep "biased" only for cell labels). Use `ignorable` (not "non-informative") for the MAR component to avoid collision with "noninformative prior". Replace colloquial `estimand drift` -> "bias in the estimand" and `design surface area` -> a defined phrase. Normalise `MNAR` spelling.	See MCAR_25 flag in Section 6.
10	Choose one orthography for `mis-specified`/`misspecified`. Introduce `cluster-robust (sandwich) standard error` once, then one term. Keep `working correlation structure` for the GEE object only; use `working covariance` for the LME structure. Fix participant noun (D7).	Most careful paper; fixes are orthographic.

4. Tier 3 - glossing / definition additions

Add a one-line first-use gloss (no statistical change) for load-bearing terms absent from the lexicon: `subadditivity` (03, 01), `frailty` (03), `balanced-placebo / open-hidden design` (06), `vulnerability window` and `design surface area` (09), `lasagna plot` (02). Consider adding `subadditivity`, `balanced-placebo design`, and `gating function` as new entries to `docs/29-...-lexicon.md` so future papers inherit a fixed definition.

5. Cross-file consistency (full vs slim)

For papers 01, 02, 06, run a final diff pass so the slim variant uses the same canonical forms and acronym expansions as the full report (e.g. IL-6 is defined in 01 full but not slim; "RM-ANOVA" vs "repeated-measures analogue" differs across 02 full/slim).

6. Out of scope: substantive issues to flag to the author

These are NOT terminology and must not be auto-edited; each needs author judgement because a "fix" could change meaning or a stated result.

1. **Paper 02, `report.Rmd` Conclusion #1.** States A2 is "the recommended default … across the full factorial", which conflicts with the paper's own data (A2 leads in ~19% of cells) and with the slim variant's correctly hedged conclusion. (Consistent with the prior project note that the A2-dominance headline is not universal.)
2. **Paper 06, Study B/C naming divergence.** The full report labels the belief-decoupling study "Study B" (sec 7.5) and the coupling pilot the "contaminated-biomarker pilot" (7.4); the slim report calls them "Study C" and "Study B pilot". Any reader cross-referencing the two files is misdirected. Decide one scheme and apply to both.
3. **Paper 09, `MCAR_25` cell label.** Its flat dropout component scales with elapsed time (a design variable), which is MAR, not MCAR, under the lexicon definitions. Renaming would change a stated result label, so confirm the mechanism before any change.

7. Execution workflow

1. Lock decisions D1-D7 (Section 1).
2. Apply Tier 1 globally, one file at a time, via the Edit tool with `replace_all`; triage D5 ("washout") hit-by-hit. Do not use `sed -i` on these cloud paths.
3. Apply Tier 2 per paper; Tier 3 glosses; Section 5 full-vs-slim diff.
4. Re-render each changed manuscript with the appropriate R environment (note: paper 04 requires the Framework R / kableExtra environment per the project render notes) and confirm it still builds and that the PDF render-timestamp footer is updated.
5. Address Section 6 items separately with the author.

Effort estimate. Tier 1 is mechanical and dominated by D2/D3/D4 (~235 substitutions) plus contextual D5 triage (~92 hits). Tier 2 is a few dozen targeted edits per paper. The re-render and full-vs-slim diff are the main time cost. None of the work changes a number, estimand, or model.

8. Execution record (2026-07-30)

Tier 1 applied across 21 files by a chunk-aware transform (prose rules skip fenced R chunks so that ggplot's `colour=` and dplyr's `summarise()` are untouched;

display-string rules such as `MC SE` are applied inside chunks as well, because those strings are table headers):

- D2 Type I/II casing, D3 `biomarker-by-treatment` -> `biomarker-treatment`, D4 `MC SE/MC SEs` -> `MCSE/MCSEs`, Tier 1.8 `OLBDC` -> `OL+BDC`.
- D1 in the US direction: `modelling`, `generalis*`, `behaviour`, `characteris*` (guarding `characteristic`), `normalis*`, `summaris*` (prose only), `artefact`, `randomis*`, `standardis*`, `dichotomis*`, `ageing`, `sceptic*`, `favour*`, `enrolment`, `signalling`, `programme`, `labelled`, `modelled`, `analyse`, `centre*`, `colour` (prose only), `whilst`, `amongst`.
- 16 of 21 files changed; residual British-looking tokens are all R identifiers (`dplyr::summarise`) and are correct as they stand.

Tier 1 items already satisfied: HTE is expanded at first use in 01 and 06; the acronym sweep (DGP, MVN, LME, MCAR/MAR/MNAR, OL/BDC, CO, PK/PD, BLUP, SEM, RM-ANOVA, OFAT, GEE, ICC) found only five true misses, fixed here: MCAR and MAR in 02, BLUP in 04, DGP and MVN in 11, plus RM-ANOVA in 01-slim and MCAR/MAR in 02-slim for Section 5 parity. ADEMP is treated as a named, always-cited framework and is not expanded.

D5 (`washout`) triage. 35 hits reviewed individually. All but three are genuine inserted clearance gaps and were kept. The three changed are in paper 02's toy example, where the span is the carryover-contaminated post-discontinuation window: `three-week washout` -> `three-week off-drug period`, `the second washout week` -> `the second off-drug week`, `throughout the washout` -> `throughout the off-drug period`.

Tier 2, applied: 02, one residual `matched-decay (M2)` reference to the specification changed to `exposure-weighted (M2)` (the matched-decay *cell* retains the term, as the plan specifies).

Tier 2, already converged: 06 (`lumped` survives only as the math superscript $\beta_{bm}^{\text{lumped}}$, not as prose); 07 (`modGompertz` absent from narrative; lowercase `gompertz` occurs only in citation keys, file paths, and equation labels); 08 (`main effect` absent; `anti-conservative` already single-form); 09 (`attrition`, `estimand drift`, `design surface area`, `non-informative` all absent; `biased dropout` survives only as cell-label description, which the plan permits); 10 (`misspecified` already uniform, 13/13).

Tier 3 glosses added: `subadditivity` (03), `frailty` (03), `balanced-placebo design` (06). `lasagna plot` has no occurrences.

Not done: the plan's suggestion to add `subadditivity`, `balanced-placebo design`, and `gating function` to `docs/29-...-lexicon.md`; Section 6 substantive items; and the second terminology pass over the `rgt` blocks, which remain placeholders.

Interaction with the notation pass. This execution followed the symbol-level pass recorded in `analysis/report/NOTATION.md`. The analysis specifications are now M1/M2/M3 in all reader-facing text, which is why this file refers to M2 where the 2026-06-17 draft said A2.

Rendered on 2026-06-17 at 19:37 PDT; execution record appended 2026-07-30. Source: ~/prj/res/36-pmsimstats-ng/pmsimstats-ng/docs/30-terminology-consistency-plan.md